Introduction to Applied Cryptography – LDACS1310
Exercises Session 6

Questions

Question 1: Diffie-Hellman Problem: too many small subgroups

The aim of this exercise is to use a simple practical example to illustrate the problem of choosing $p = 2q_1 \cdots q_l + 1$ with small distinct primes $2, q_2, \ldots, q_l$. Your task will be to find the secret key of alice x from the public the public $X = g^x$. In this exercise, we fix $l = 3$. We have $p = q + 1$, and $q = q_1 q_2 q_3$ with the prime $q_1 = 2$ to obtain $p = q + 1$.

Let have g, a generator such that $|\langle g \rangle| = q = q_1 q_2 q_3 = p - 1$. Then $g_i = g^{q_1 \cdots q_{i-1} q_{i+1} \cdots q_l} = g^{q/q_i} \bmod p$ is of order q_i. In our example we have $g_1 = g^{q_2 q_3} \bmod p$, $g_2 = g^{q_1 q_3} \bmod p$, and $g_3 = g^{q_1 q_2} \bmod p$.

Let $x \in \{0, \ldots, q-1\}$ be the secret, and $X := g^x \bmod p$. We can decompose X into (X_1, X_2, X_3) with smaller exponents (x_1, x_2, x_3) with respective basis (g_1, g_2, g_3). We simply have $X_i = X^{q/q_i} \bmod p$ which satisfies $X_i = (g^{q/q_i})^x = g_i^x = g_i^{x_i} \bmod p$, with $x_i = x \bmod q_i$.

Assuming we compute (x_1, x_2, x_3) from (smaller) discrete log extractions, we can reconstruct $x = \epsilon_1 x_1 + \epsilon_2 x_2 + \epsilon_3 x_3 \bmod q$, from $\epsilon_1, \epsilon_2, \epsilon_3 \in \mathbb{Z}_q$, where

$$\epsilon_1 = \begin{cases} 1 \bmod q_1 \\ 0 \bmod q_2 \\ 0 \bmod q_3 \end{cases} \quad \epsilon_2 = \begin{cases} 0 \bmod q_1 \\ 1 \bmod q_2 \\ 0 \bmod q_3 \end{cases} \quad \epsilon_1 = \begin{cases} 0 \bmod q_1 \\ 0 \bmod q_2 \\ 1 \bmod q_3 \end{cases}$$

Indeed, $x \bmod q_i = ([\epsilon_1 \bmod q_i] \cdot x_1 + [\epsilon_2 \bmod q_i] \cdot x_2 + [\epsilon_3 \bmod q_i] \cdot x_3) \bmod q_i = x_i$, as long as $\epsilon_1 = q_2 q_3 [(q_2 q_3)^{-1} \bmod q_1]$, $\epsilon_2 = q_1 q_3 [(q_1 q_3)^{-1} \bmod q_2]$, and $\epsilon_3 = q_1 q_2 [(q_1 q_2)^{-1} \bmod q_3]$.

Let have $p = 443$, $q = 442$, $g = 2$, and $X = 84$.

1. Find q_1, q_2, and q_3
2. Compute g_1, g_2, and g_3
3. Compute ϵ_1, ϵ_2, and ϵ_3
4. Compute x_1, x_2, x_3, and then x
5. Choose $y \in \{0, \cdots, q-1\}$, compute $Y = g^y \bmod p$ and repeat step 4. Note that you need to find your y.